IT314 SOFTWARE ENGINEERING | GROUP 27

Software Design Diagram Generator

(SDDG)

Software Requirements Specification

Sprint 1 Baseline

1. Functional Requirements

Requirement text follows SRS v2.0 [R1]. Where the two editions of v2.0 differ, the reconciliation is recorded in Appendix C.

1.1 Requirement and Document Ingestion

ID	Requirement
FR1	The system shall accept requirements through these input methods: typed or pasted text; uploaded PDF or DOCX SRS documents (maximum 10 MB per file); existing user stories; and existing FR/NFR lists. Input language is English.
FR2	The system shall validate every input before processing. It shall reject, with a specific error message stating what is wrong, any file that exceeds the size limit, has an unsupported type, is corrupted, is password-protected, or is a PDF with no extractable text on any page. A PDF that has both text pages and image-only pages shall be accepted: only the text pages are extracted and the user is warned which pages were skipped. The system shall never generate content from an unreadable page and shall never fail silently.
FR3	The system shall let users create, name, rename, list, switch between and delete projects. Project names are unique per owner (case-insensitive). Within a project, users shall be able to upload multiple documents and combine typed or pasted text, uploaded files, user stories and FR/NFR lists, up to a maximum of 15 documents or 100 total pages per project, whichever limit is reached first.

1.2 Requirement Analysis and Information Extraction

ID	Requirement
FR4	The system shall split all input into individually identified requirement statements (each with a unique ID and source location) and extract from them the modelling information: actors, entities/classes, processes, data flows, data stores, interactions and relationships between elements. Each extracted item shall link to its source requirement statement(s).
FR5	Before any diagram is generated, the system shall let the user view, edit, add, delete and explicitly confirm the extracted information. Any later change clears the confirmation. Generation shall be impossible without current confirmation, enforced on the server.

1.3 Ambiguity and Requirement Validation

ID	Requirement
FR6	The system shall detect ambiguous, incomplete, inconsistent and contradictory requirements and record each finding as a flag on the affected requirement(s).
FR7	For each flag the system shall highlight the requirement, explain in plain language why it was flagged, and either ask the user a clarifying question or let the user resolve it manually, with a suggested fix where possible.
FR8	The system shall still allow diagrams to be generated with unresolved flags (there is no blocking severity), and the warnings for unresolved flags shall remain visible on the diagrams.

1.4 Diagram Recommendation and Generation

ID	Requirement
FR9	The system shall analyse the requirements to recommend suitable diagram types with a reason for each, let the user add or remove types, and require the user to confirm the final selection before generation starts.

ID	Requirement
FR10	The system shall generate Use Case, Class, Activity, Sequence, DFD and Component diagrams (delivered progressively per the sprint plan) from the confirmed requirements, and shall show a progress or status indicator during generation. Start and end times of every run shall be recorded.
FR11	The system shall notify the user in-app, and optionally by email (off by default), when a generation or regeneration completes or fails, including when the user has left the project.

1.5 Diagram Review and Editing

ID	Requirement
FR12	The system shall present every generated diagram as an editable draft supporting adding, removing, renaming and moving elements, changing relationships, and editing diagram-specific properties. Edits are held in the editor until the user saves; each save creates a new version.
FR13	The system shall support regenerating either a selected portion or the entire diagram. Before an entire diagram is regenerated, its current state shall be saved as a version.
FR14	The system shall allow users to modify source requirements, tell them which diagrams are affected, and regenerate the affected diagrams after the user approves (FR17). It shall keep a version history of each diagram (latest 20 versions) so users can view or revert to a previous version without losing newer ones.
FR15	The system shall allow users to customise the visual presentation of a diagram (colour scheme, layout direction, font size) before export, without altering its structural content.

The editing mechanism for FR12 (in particular “moving” elements) depends on open decision OD-2.

1.6 Traceability and Consistency

ID	Requirement
FR16	The system shall maintain a single shared requirements model behind all generated diagrams. Diagram elements reference model items rather than holding copies, so a change to the model is reflected in every dependent diagram’s traceability view.
FR17	The system shall explain how each diagram element was derived from the requirements; flag any requirement not represented in any diagram; prompt the user to propagate a requirement change to the diagrams it affects (declining leaves them marked “out of date”); warn the user when a manual diagram edit drifts from the current requirements; and flag shared elements (actors, entities, processes) that are named inconsistently across diagrams of the same project, matching by model item ID and offering to apply one name everywhere.

1.7 Export and Documentation

ID	Requirement
FR18	The system shall allow diagrams to be exported as PNG, SVG and PDF (A4) with consistent page sizing and readable labels, and from Sprint 3 as editable draw.io XML and as PlantUML text. It shall provide a one-click export using the user’s default format (PNG unless changed).

1.8 User Guidance

ID	Requirement
FR19	The system shall guide the user through the full pipeline, from raw requirements to a reviewed and finalised diagram (guidance can be dismissed permanently), and shall offer contextual explanations for any diagram notation element.

1.9 Project Management and Reporting (Owner-only)

ID	Requirement
FR20	The system shall provide a project-level summary showing requirement coverage: how many requirements have generated diagrams, how many have unresolved flags (FR6 to FR8), and how many are not represented in any diagram (FR17).
FR21	The system shall allow a project Owner to export a combined project report (requirements, diagrams and open issues) as a single PDF suitable for sharing outside the tool.
FR22	The system shall log key project activity (who changed a requirement, who edited, saved, regenerated or reverted a diagram, and when) so that Owners can review recent changes.

1.10 Accounts, Access and Administration

ID	Requirement
FR23	The system shall allow a System Administrator to view, add, deactivate and reactivate user accounts. Deactivation revokes access but does not delete the user's projects.
FR24	The system shall allow a project Owner or a System Administrator to invite collaborators and assign each a role (Viewer, Editor or Owner, Section 2.4), and allow Owners to remove collaborators. Roles shall be enforced on the server and take effect on the next request. The system shall detect when a save is based on an outdated revision and warn the user instead of overwriting.
FR25	The system shall allow a System Administrator to view system usage and configuration: active user count, total projects, storage used against defined limits, recent error and failure logs, generation-time statistics against the NFR1 target, and the current limits and settings (read-only, no secrets).

1.11 Search and Navigation

ID	Requirement
FR26	The system shall allow users to search and filter requirements and diagrams within a project by keyword, ID, status (flagged, unresolved, exported) or diagram type. Filters of different categories combine with AND; values within one category combine with OR.

1.12 User Feedback

ID	Requirement
FR27	The system shall let users report issues or give feedback on a specific diagram or requirement inside the tool. Owners see the reports on their projects and administrators see all reports.

1.13 Additional Technical Features

ID	Requirement
FR28	The system shall let users create an account by signing up with email and password (email verification required; password of at least 8 characters with a letter and a digit) or by signing in with Google; log in and out; and reset a forgotten password by a single-use link valid for 1 hour. Five failed logins lock the account for 15 minutes.
FR29	After generation the system shall validate each diagram against its notation rules (DR1), against the confirmed extracted items and for structural problems, show the resulting warnings and suggestions on the diagram, and let the user either edit the diagram or accept a warning as-is.
FR30	The system shall handle external-service problems gracefully: on AI-provider rate limits it shall queue and automatically retry (up to 3 retries within 5 minutes); on failure it shall show a plain-language message with a Retry option and shall not lose the user's inputs or edits; and when the backend is starting up it shall tell the user instead of showing an error.

1.14 Technical Enablers

ID	Enabler
TE1	Source repository (GitHub monorepo) with continuous-integration actions.
TE2	Automated deployment of frontend and backend, with an uptime monitor on the health route.
TE3	Database and file storage with versioned schema migrations.
TE4	AI-service module: provider, model and keys are configuration; keys are never sent to the browser; logs contain metadata only, never document content; retry and failure reporting hooks.
TE5	Transactional email integration.
TE6	Background scheduled-job runner for clean-up, audit maintenance and asynchronous tasks.

2. Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	For a typical document (Section 1.4), generation of the selected diagram types shall have a median of at most 60 seconds and a 95th percentile of at most 120 seconds. Runs over 120 seconds are recorded and shown to administrators.
NFR2	Usability	A first-time user shall be able to complete upload, generation and export unaided within 1 hour. Verified in a test with at least 5 first-time users, of whom at least 4 succeed.
NFR3	Security and privacy	Uploaded documents and extracted project data shall be encrypted at rest (AES-256 via the storage or database provider) and in transit (HTTPS/TLS only).
NFR4	Security and privacy	The system shall authenticate users (sessions expire after 24 hours of inactivity) and enforce access control per project on the server. Requests for projects the user cannot access shall return “not found”.
NFR5	Transparency	The system shall provide an in-app Privacy Policy stating how documents are processed (including that content is sent to an external AI provider), stored and deleted, and shall obtain and record the user’s consent before first AI processing. Data of an inactive account is permanently deleted after 30 days (warning email between day 25 and 30) or immediately on the user’s request. Deletion removes the account, documents, diagrams, versions and feedback; sole-owned projects with recently active Editors are transferred to the most recently active Editor instead of deleted; only anonymised, content-free audit records are kept, for 90 days.
NFR6	Maintainability and efficiency	The system shall support incremental regeneration so that a change affects only the relevant diagram elements. A change to one requirement statement (affecting at most 10 diagram elements) shall complete within 30 seconds.
NFR7	Availability	The system shall be available during weekday operating hours (Monday to Friday, 08:00 to 20:00 IST) with at least 95% monthly availability. Downtime is allowed on weekends; maintenance can be scheduled only on Saturday or Sunday.
NFR8	Compatibility	The system shall work correctly on desktop Chrome, Firefox and Edge, each on its latest two stable major versions, at window widths of 1280 px and above.
NFR9	Scalability	The system shall support at least 5 concurrent users without violating NFR1; further simultaneous generation requests are queued in order.
NFR10	Security and privacy	The system shall support collaborative workspaces in which access to a shared project is governed by the roles assigned under FR24.

3. Domain Requirements

ID	Requirement
DR1	The system shall model software using recognised software-engineering notations, each following its own standard conventions: UML Use Case, Class, Activity, Sequence and Component diagrams, and Data Flow Diagrams (processes, data flows, data stores and external entities as distinct symbols).
DR2	Diagram content shall be built only from requirement-derived actors, entities, processes, interactions and relationships. Where a requirement does not give enough information for an element, the system shall mark that element as an assumption, visually distinguish it from requirement-derived elements, and preserve that marking in every export format.